INHERITANCE GENETICS

Module map

- Module 09: Inheritance Genetics
- Connected lab: lab-10_inheritance-genetics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 09
Inheritance Genetics

1. Alleles and genotype
2. Mendelian patterns
3. Punnett-square prediction
4. Beyond simple dominance
5. Pedigrees and probability

Lab evidence
lab-10_inheritance-genetics.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

INHERITANCE GENETICS

Learning objectives

- Distinguish gene, allele, genotype, and phenotype.
- Use segregation to explain monohybrid inheritance.
- Build and interpret Punnett-square probabilities.
- Recognize inheritance patterns beyond complete dominance.
- Use pedigree evidence to reason about traits.

OBJECTIVE 1
Distinguish gene, allele, genotype, and phenotype.

OBJECTIVE 2
Use segregation to explain monohybrid inheritance.

OBJECTIVE 3
Build and interpret Punnett-square probabilities.

OBJECTIVE 4
Recognize inheritance patterns beyond complete dominance.

OBJECTIVE 5
Use pedigree evidence to reason about traits.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

INHERITANCE GENETICS

Topic sequence

- Alleles and genotype: Alleles are gene variants that contribute to genotype and phenotype.
- Mendelian patterns: Mendelian inheritance predicts patterns when alleles segregate into gametes.
- Punnett-square prediction: Punnett squares model probability, not guaranteed outcomes.
- Beyond simple dominance: Incomplete dominance, codominance, polygenic traits, and environment complicate simple patterns.
- Pedigrees and probability: Pedigrees use family evidence to infer inheritance patterns.

01

Alleles and genotype

Alleles are gene variants that contribute to genotype and phenotype.

02

Mendelian patterns

Mendelian inheritance predicts patterns when alleles segregate into gametes.

03

Punnett-square prediction

Punnett squares model probability, not guaranteed outcomes.

04

Beyond simple dominance

Incomplete dominance, codominance, polygenic traits, and environment complicate simple patterns.

05

Pedigrees and probability

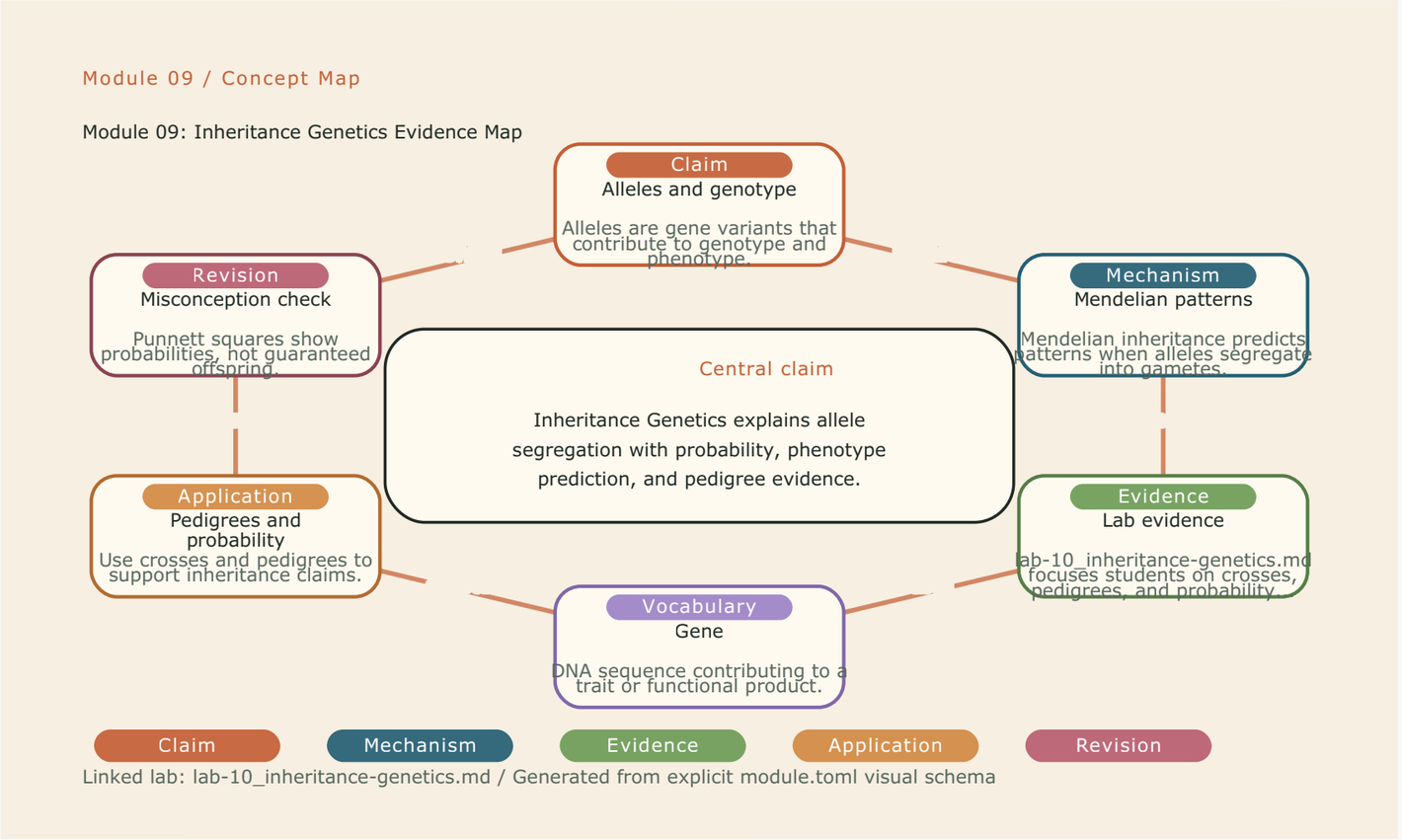
Pedigrees use family evidence to infer inheritance patterns.

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

INHERITANCE GENETICS

Module 09: Inheritance Genetics Evidence Map

- Module focus: Inheritance Genetics
- Central claim: Inheritance Genetics explains allele segregation with probability, phenotype prediction, and pedigree evidence.
- Key nodes to connect: Alleles and genotype, Mendelian patterns, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-10_inheritance-genetics.md supplies an evidence surface for the map.

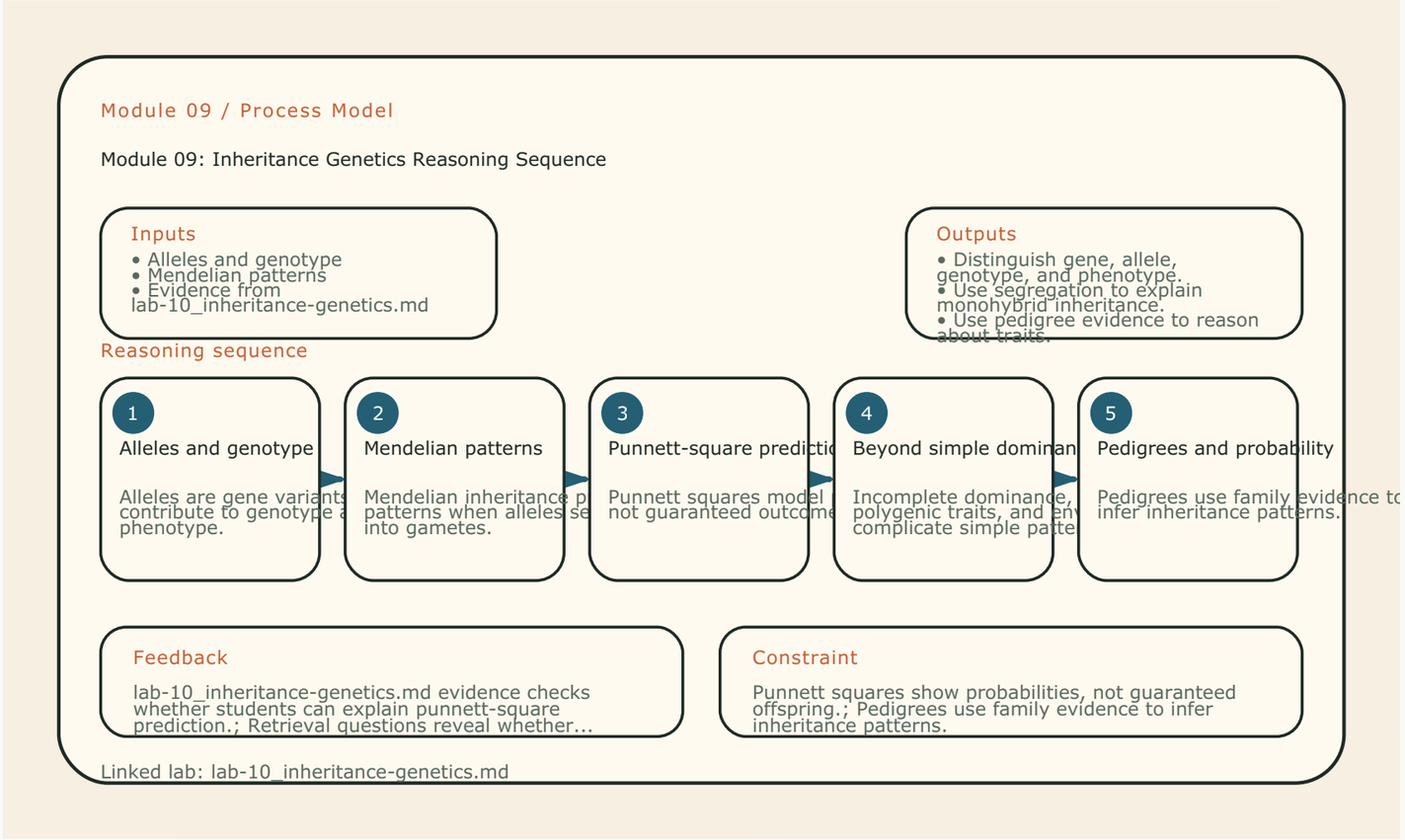


Teaching note: Ask students to explain one edge in their own words before moving on.

INHERITANCE GENETICS

Module 09: Inheritance Genetics Reasoning Sequence

- Module focus: Inheritance Genetics
- Trace the model: Alleles and genotype -> Mendelian patterns -> Punnett-square prediction -> Beyond simple dominance
- Inputs become outputs: Alleles and genotype, Mendelian patterns -> Distinguish gene, allele, genotype, and phenotype., Use segregation to explain monohybrid inheritance.
- Feedback check: lab-10_inheritance-genetics.md evidence checks whether students can explain punnett-square prediction.
- Lab connection: lab-10_inheritance-genetics.md gives students a process to observe or test.



Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

INHERITANCE GENETICS

Terms as evidence handles

- Gene: DNA sequence contributing to a trait or functional product.
- Allele: Variant form of a gene.
- Genotype: Allele combination of an organism.
- Phenotype: Observable trait or characteristic.
- Dominant: Allele effect visible with one copy in simple dominance.
- Recessive: Allele effect masked by dominant allele in heterozygotes.

GENE
DNA sequence contributing to a trait or functional product.

ALLELE
Variant form of a gene.

GENOTYPE
Allele combination of an organism.

PHENOTYPE
Observable trait or characteristic.

DOMINANT
Allele effect visible with one copy in simple dominance.

RECESSIVE
Allele effect masked by dominant allele in heterozygotes.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

INHERITANCE GENETICS

Lab connection

- Lab file: lab-10_inheritance-genetics.md
- Lab evidence should test or illustrate: Punnett-square prediction
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Alleles and genotype

EVIDENCE
lab-10_inheritance-genetics.md

CLAIM
Distinguish gene, allele, genotype, and phenotype.

Teaching note: Make the lab connection explicit before students encounter the activity.

INHERITANCE GENETICS

Contrast check

- Do not stop at: Alleles are gene variants that contribute to genotype and phenotype.
- Stronger explanation: Pedigrees use family evidence to infer inheritance patterns.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Alleles are gene variants that contribute to genotype and phenotype.

DEEPER

Pedigrees use family evidence to infer inheritance patterns.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

INHERITANCE GENETICS

Module 09: Inheritance Genetics Retrieval and Lab Check

- Module focus: Inheritance Genetics
- Retrieval prompt: How are genes and alleles related?
- Answer check: Distinguish gene, allele, genotype, and phenotype.
- Required terms: Gene, Allele, Genotype
- Lab connection: lab-10_inheritance-genetics.md; lab-10_inheritance-genetics.md supplies evidence for crosses, pedigrees, and probability patterns in inheritance.

Module 09 / Retrieval Card

Module 09: Inheritance Genetics

Retrieval and Lab Check

Routine

Cover notes -> answer aloud -> cite evidence -> revise

1 Cover notes 2 Answer aloud 3 Cite evidence 4 Revise

Q1

How are genes and alleles related?

Check: Distinguish gene, allele, genotype, and phenotype.

Q2

How do genotype and phenotype differ?

Check: Use segregation to explain monohybrid inheritance.

Q3

What does heterozygous mean?

Check: Build and interpret Punnett-square probabilities.

Q4

How does allele segregation explain a monohybrid cross?

Check: Recognize inheritance patterns beyond complete dominance.

Terms to use

Gene, Allele, Genotype, Phenotype, Dominant

Lab connection

lab-10_inheritance-genetics.md supplies evidence for crosses, pedigrees, and probability patterns in...

Intro biology routine: claim, evidence, reasoning, revision.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

INHERITANCE GENETICS

Practice quiz bridge

- 1. An organism's genotype is its:
- 2. Two heterozygotes are crossed (Aa x Aa). What fraction of offspring are expected to show the recessive trait?
- 3. A heterozygote (Aa) shows the dominant trait because:
- 4. What is a pedigree used for?

QUESTION 1

Key: C

An organism's genotype is its:

QUESTION 2

Key: A

Two heterozygotes are crossed (Aa x Aa). What fraction of offspring are expected to show the recessive trait?

QUESTION 3

Key: D

A heterozygote (Aa) shows the dominant trait because:

QUESTION 4

Key: B

What is a pedigree used for?

Teaching note: Use the quiz as formative feedback. Answer key: 1=C, 2=A, 3=D, 4=B

INHERITANCE GENETICS

Synthesis and exit ticket

- Exit claim: explain alleles and genotype using evidence from lab-10_inheritance-genetics.md.
- Must include: Gene, Allele, Genotype.
- Revision prompt: what evidence would change your explanation?

CLAIM
Alleles and genotype

EVIDENCE
lab-10_inheritance-genetics.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.